AP Project Report: University ERP

CSE201: Advanced Programming

Tech Stack: Java Swing, JDBC, MySQL, JUnit

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1. Introduction

1.1 Project Overview

The University ERP System is a robust desktop application designed to digitize and streamline the academic operations of a university. The system acts as a centralized platform connecting three key stakeholders: Administrators, Instructors, and Students. It handles complex workflows such as course scheduling with deadlines, conflict-free student registration, grading, and data archival, ensuring data integrity through a dual-database architecture.

1.2 Key Features

- **Role-Based Access Control (RBAC):** Distinct dashboards for Admin, Instructor, and Student, ensuring users only access authorized data.
- **Dual Database Architecture:** Strict separation of concerns using `auth_db` for secure credentials and `erp_db` for academic records.
- **Smart Maintenance Mode:** A global control that instantly restricts write operations (registration, grading) across all active sessions without requiring a restart.
- **Advanced Scheduling:** Implementation of registration/drop deadlines and logic to prevent time conflicts or duplicate course enrollments.
- **Data Portability:** Features for Backup/Restore of the entire database and CSV export for Transcripts and Class Statistics.

1.3 Technology Stack

- **Language:** Java (JDK 17).
- **UI Framework:** Java Swing with a Custom Dark Theme.
- **Database:** MySQL 8.0 (JDBC for connectivity).
- **Security:** `jBCrypt` for password hashing.
- **Utilities:** `OpenCSV` for reporting, `Mockito/JUnit 5` for testing.

2. System Architecture

2.1 Database Schema

The system employs a "UNIX Shadow" style architecture, separating authentication from operational data.

2.1.1 Auth Database (`auth_db`) Manages login credentials and security locks.

- `users_auth`: Stores `username`, `role`, `password_hash`, and account lockout status (`failed_attempts`, `lockout_time`).

2.1.2 ERP Database (`erp_db`) Stores academic profiles and transaction data.

- `students`, `instructors`, `admins`: Profiles linked to `users_auth` via `user_id`.
- `courses`: Catalog data (`code`, `title`, `credits`).
- `sections`: Scheduling instances including `day_time`, `room`, `capacity`, and **Deadlines** (`reg_deadline`, `drop_deadline`).
- `enrollments`: Links students to sections.
- `grades`: Stores component scores and final grades.
- `settings`: Key-value store for global flags like `maintenance_on`.

2.2 Grading Logic

The system uses a fixed weighted average formula. Instructors enter raw scores, and the system computes the result:

1. **Quiz:** 20%
2. **Mid-Term:** 30%
3. **End-Sem:** 50%

Logic: The `GradeService` sums the components. If a score is missing, it is treated as pending.

2.3 Letter Grade Assignment

Final grades are calculated automatically based on the weighted sum:

- **A:** 90 - 100
- **B:** 80 - 89
- **C:** 70 - 79
- **D:** 60 - 69
- **F:** Below 60

3. Module Details

3.1 Admin Module

The Administrator has full control over system configuration and data integrity.

- **Course & Section Management:** Admins create courses and schedule sections. Uniquely, they define `Registration` and `Drop` deadlines per section.
- **User Management:** Creates users in both databases simultaneously. Supports deletion with cascade protection (preventing deletion of sections with active students).
- **System Controls:**
  - **Maintenance:** Toggles a system-wide read-only mode. Features an "Instant Refresh" capability where the banner appears immediately on the dashboard.
  - **Backup/Restore:** Allows the Admin to dump the SQL state of both databases to a file and restore them later.

3.2 Instructor Module

Instructors manage specific sections assigned to them by the Admin.

- **My Sections:** A grid view showing only courses assigned to the logged-in instructor.
- **Gradebook:** A table interface to enter raw scores. Input validation prevents negative scores or scores exceeding component limits (e.g., >20 for Quiz).
- **Analytics:** View class averages and export the performance report to CSV.

3.3 Student Module

Students use the portal to manage their academic lifecycle.

- **Registration Logic:** The system enforces strict rules:
  1. **One-Section Rule:** A student cannot register for Section B if already in Section A of the same course.
  2. **Time Conflict:** Prevents registering for sections with overlapping schedules.
  3. **Deadlines:** Blocks registration or dropping after the specific dates set by the Admin.
- **Timetable:** A generated grid showing the student's weekly schedule.
- **Transcript:** View grades and download an official CSV transcript.

4. Compilation and Execution

4.1 Prerequisites

- Java JDK 17 or higher.
- MySQL Server (Port 3306).
- Maven.

4.2 Database Configuration

Connection settings are defined in `src/main/java/edu/univ/erp/data/DBConnectionManager.java`.

- **User:** `root`

- **Password:** `“Enter your SQL pass”`

4.3 Build and Run

1. **Seed Database:** Import `auth_db_seed.sql` and `erp_db_seed.sql` into MySQL to set up the schema and default users.
2. **Build:** Run `mvn clean install`.
3. **Run:** Execute the main class `edu.univ.erp.ui.CommonWindows.LoginWindow`.

4.4 Default Credentials

- **Admin:** `admin1` / `adminpass`
- **Instructor:** `inst1` / `instpass`
- **Student:** `stu1` / `stupass`

5. Authentication and Security

5.1 Secure Authentication

Passwords are never stored in plaintext. The system uses the **BCrypt** hashing algorithm (`jBCrypt`).

- **Creation:** `BCrypt.hashpw(password, BCrypt.gensalt())`
- **Verification:** `BCrypt.checkpw(input, storedHash)`

5.2 Account Lockout

To prevent brute-force attacks, the `AuthService` tracks failed login attempts.

- After **3 failed attempts**, the account is locked for **5 minutes**.
- The login window displays a countdown timer for locked accounts.

5.3 Database Separation

The architecture strictly enforces separation:

- `AuthService` connects only to `auth_db`.
- `StudentService`/`GradeService` connect only to `erp_db`.
- The `LoginWindow` acts as the bridge, authenticating via Auth DB and then fetching the profile from ERP DB using the shared `user_id`.

6. Additional Features

6.1 Implemented Features

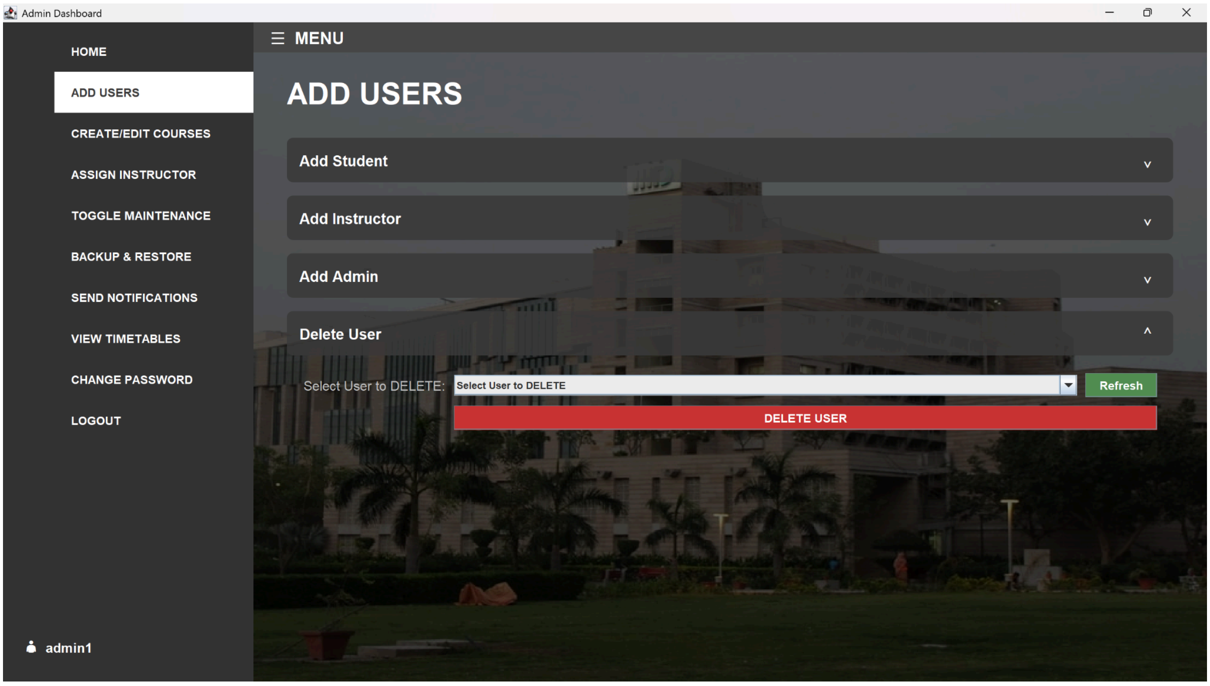
1. **Conflict Detection:** The `TimeUtil` class parses day/time strings (e.g., "Mon/Wed 10:00-11:00") to mathematically verify overlaps before allowing registration.
2. **Instant Maintenance Updates:** The Admin Dashboard communicates directly with child panels. When the maintenance toggle is switched, a callback forces the UI to repaint the "Maintenance Banner" instantly, providing immediate visual feedback.
3. **Section Management:** Includes a dedicated "Delete Section" workflow that validates against active enrollments to preserve data integrity.

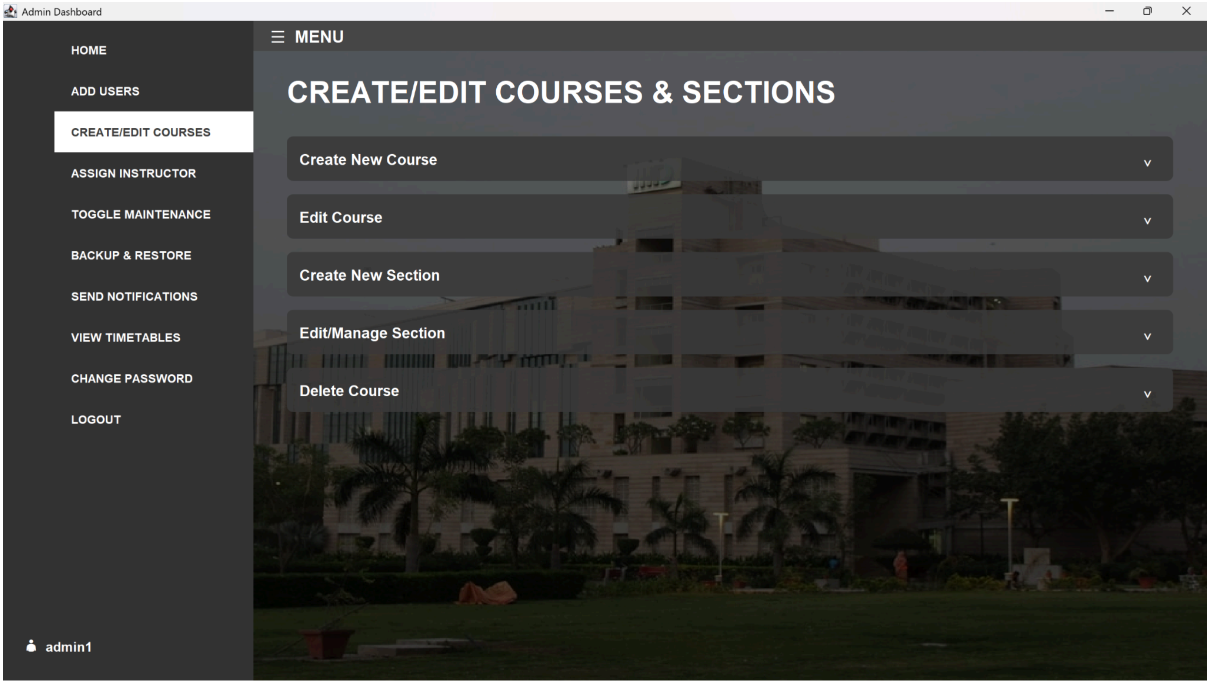
6.2 Bonus Features

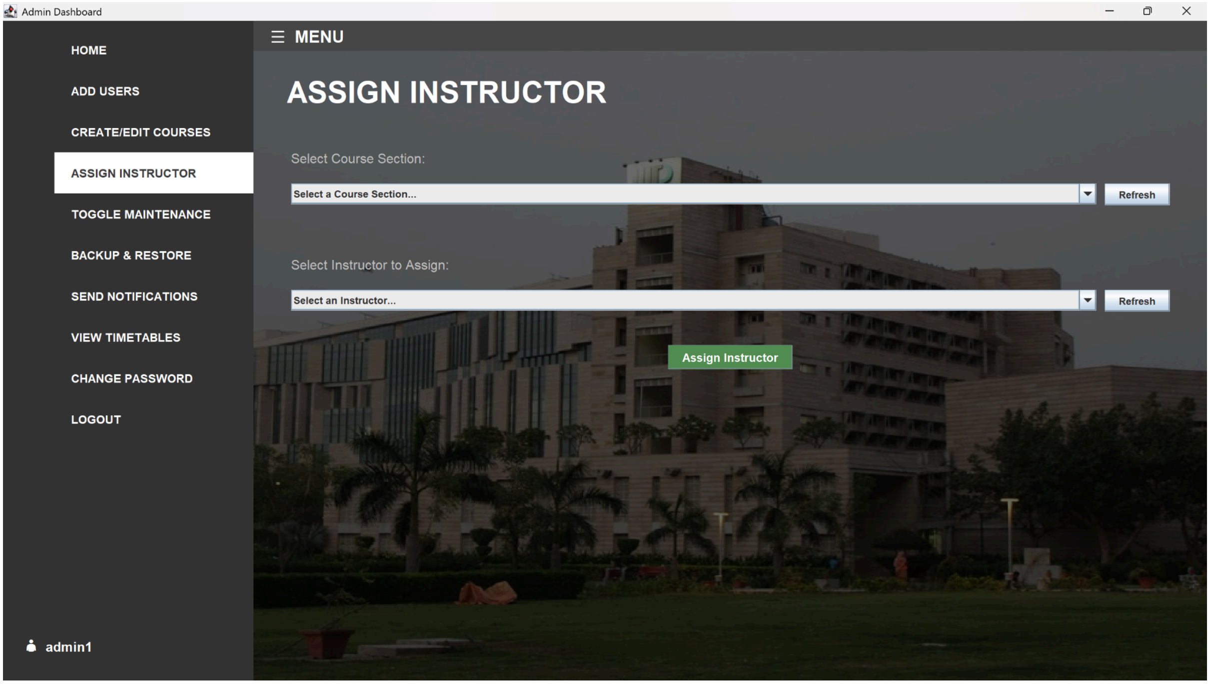
1. **CSV Export Engine:** A reusable `CsvExporter` utility allows any table data (Transcripts, Timetables, Stats) to be exported to a user-selected file location.
2. **Backup & Restore:** A dedicated Admin panel utilizes `mysqldump` and `mysql` processes to create full SQL backups of the system state, satisfying data recovery requirements.
3. **Unit Testing:** The project includes a `src/test` suite using **JUnit 5** and **Mockito** to verify core business logic (Services) in isolation from the database.

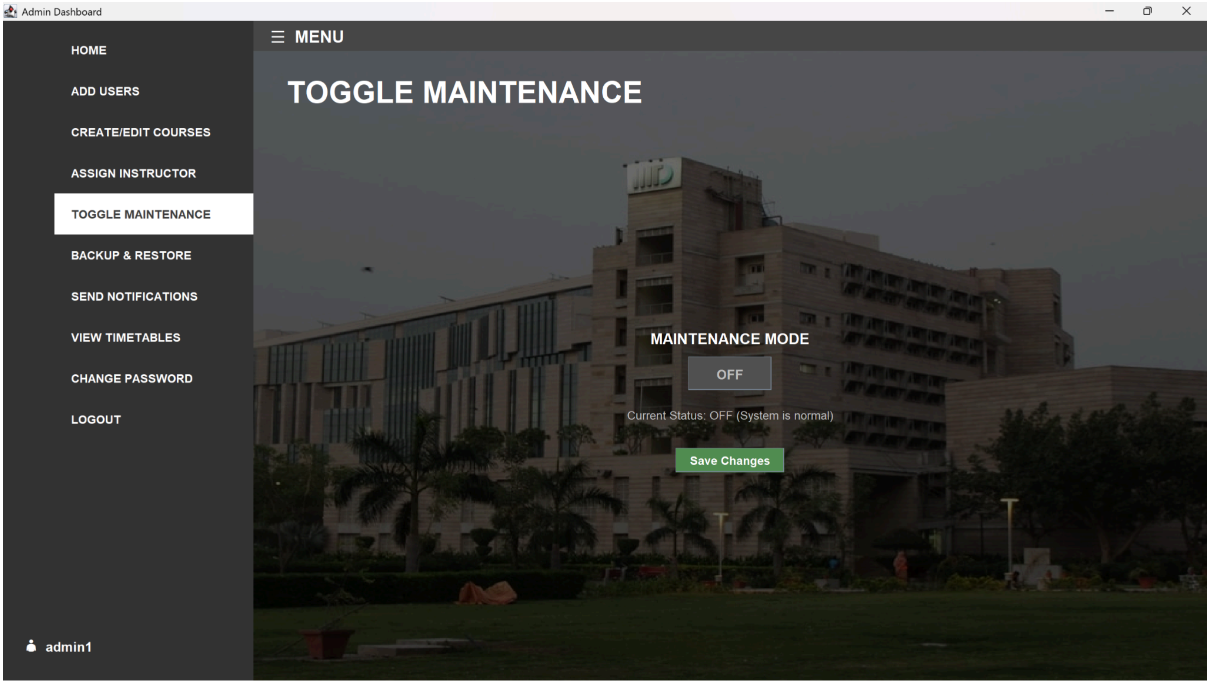
7. Project Screenshots

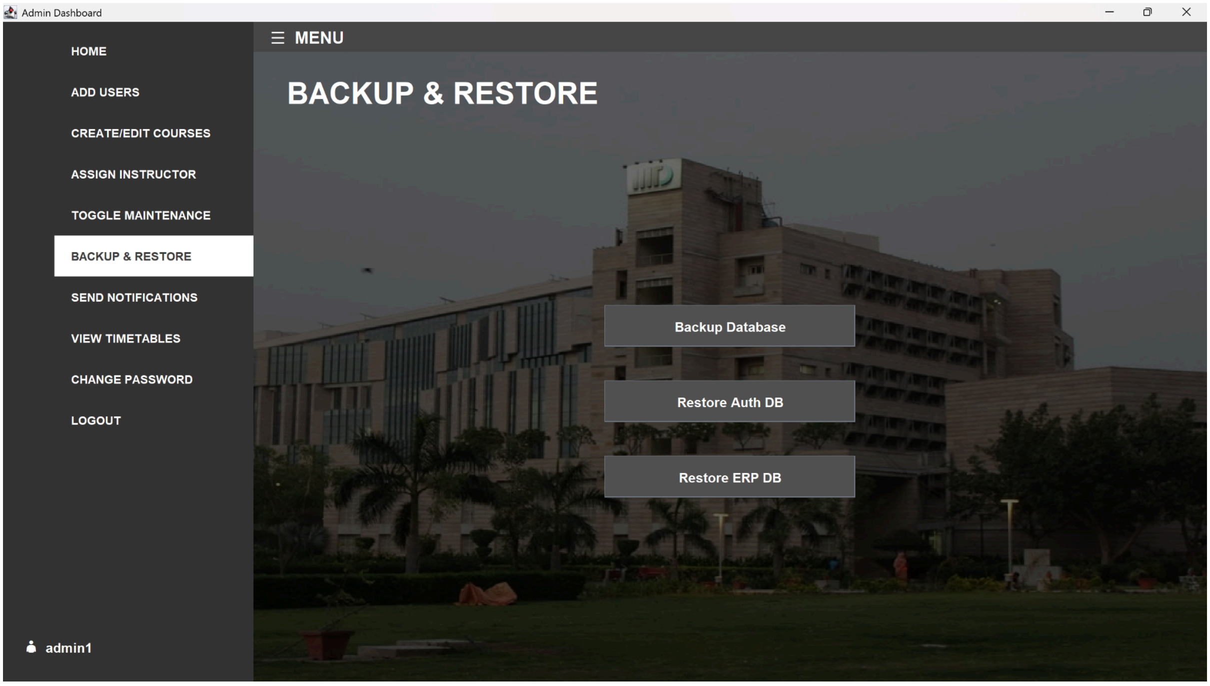
ADMIN

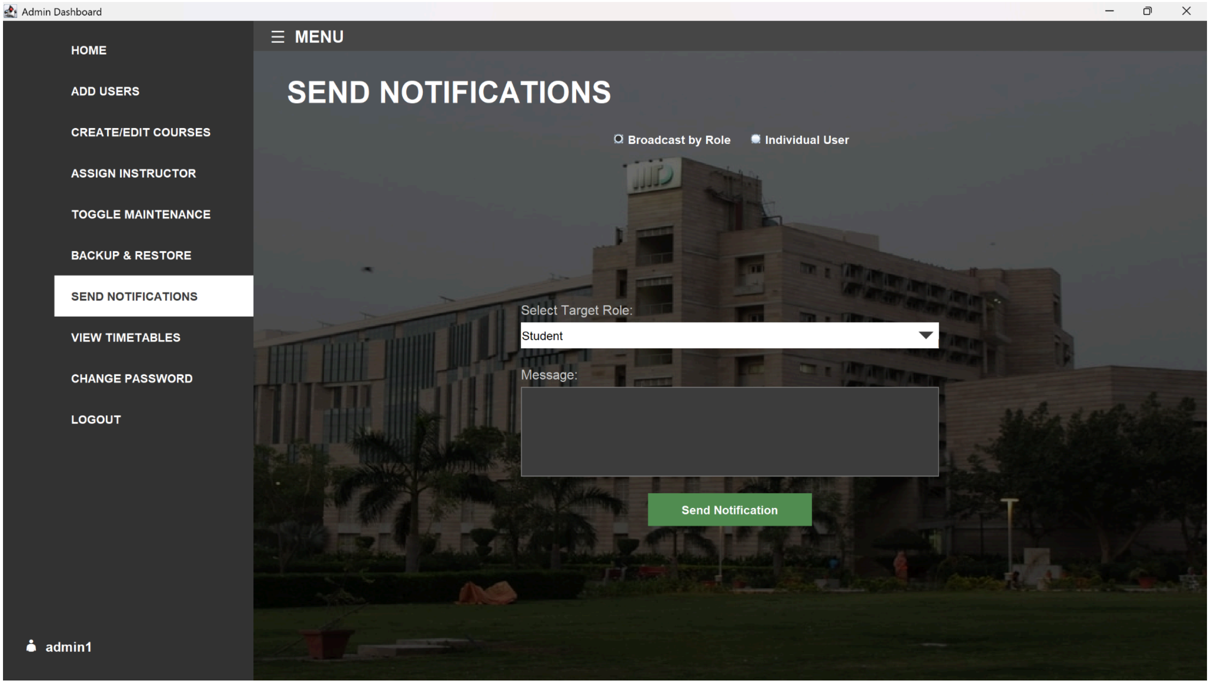


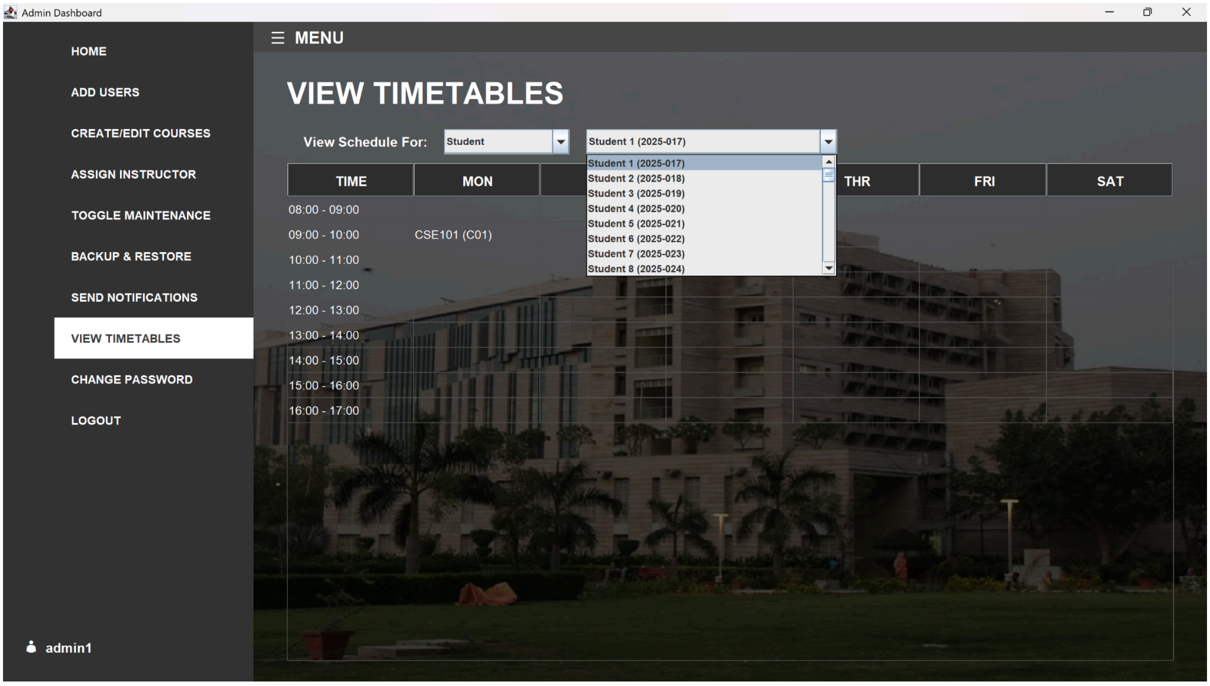

Admin Dashboard
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
MENU
CREATE/EDIT COURSES & SECTIONS
Create New Course
Edit Course
Create New Section
Edit/Manage Section
Delete Course

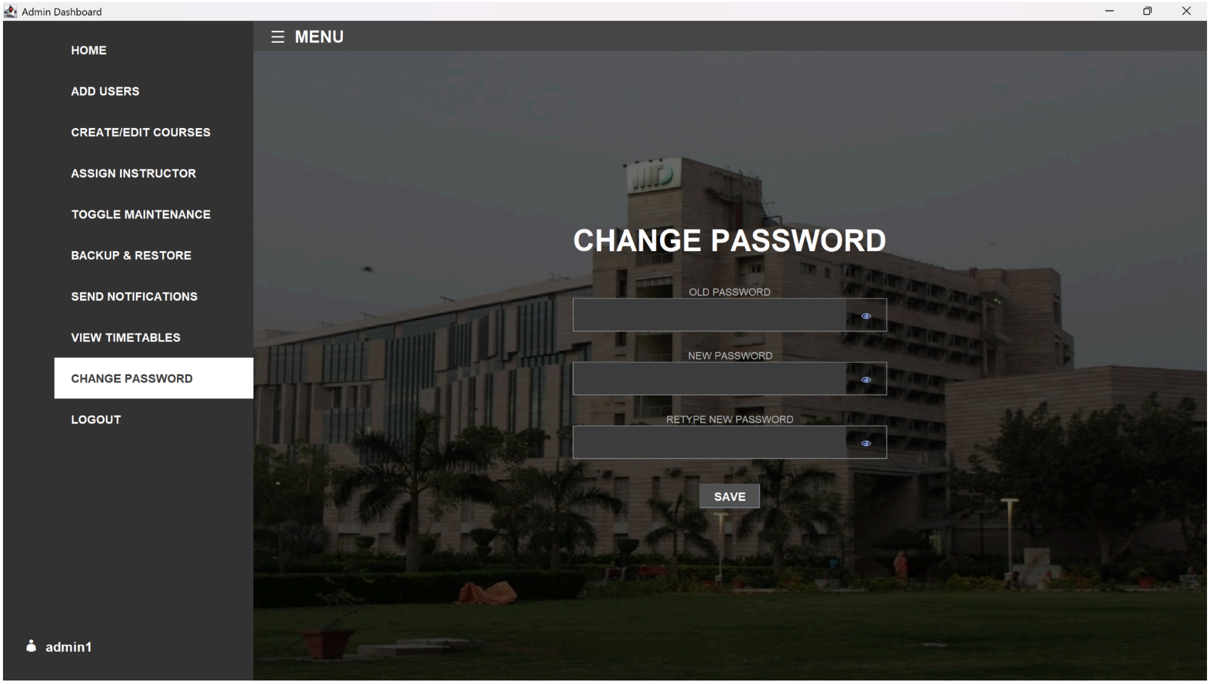

Admin Dashboard
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
MENU
ASSIGN INSTRUCTOR
Select Course Section:
Select a Course Section...
Refresh
Select Instructor to Assign:
Select an Instructor...
Refresh
Assign Instructor


Admin Dashboard
MENU
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
TOGGLE MAINTENANCE
MAINTENANCE MODE
OFF
Current Status: OFF (System is normal)
Save Changes

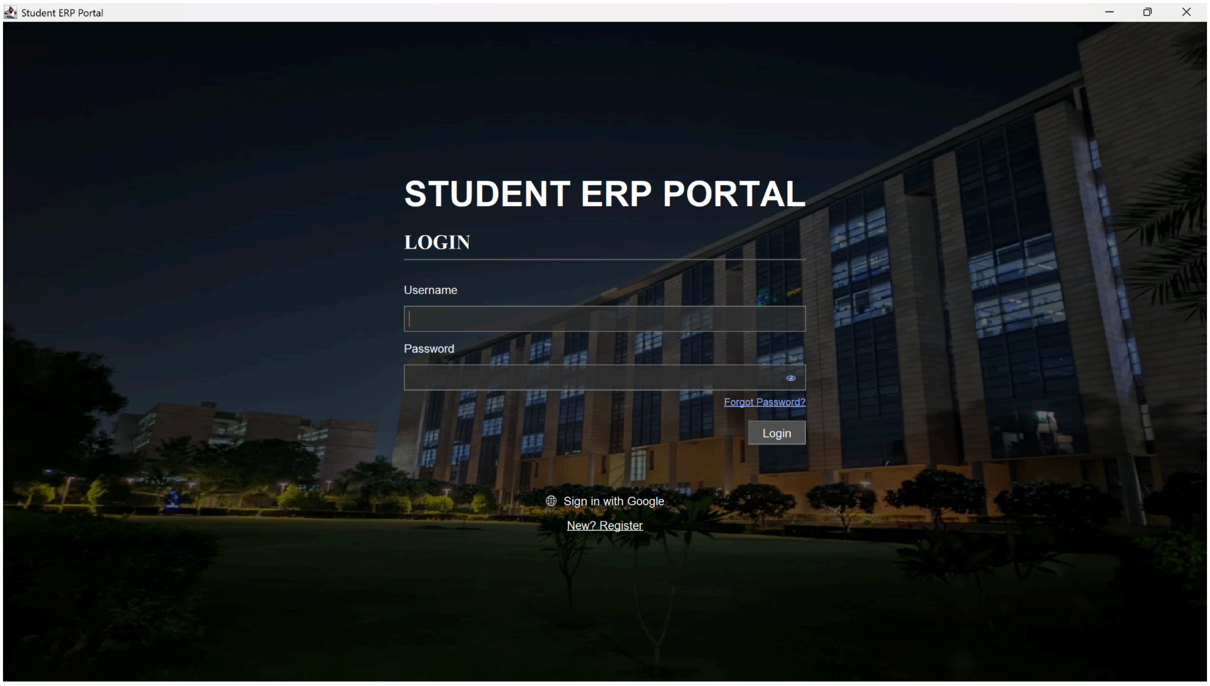

Admin Dashboard
MENU
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
BACKUP & RESTORE
Backup Database
Restore Auth DB
Restore ERP DB


Admin Dashboard
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
MENU
SEND NOTIFICATIONS
Broadcast by Role
Individual User
Select Target Role:
Student
Message:
Send Notification

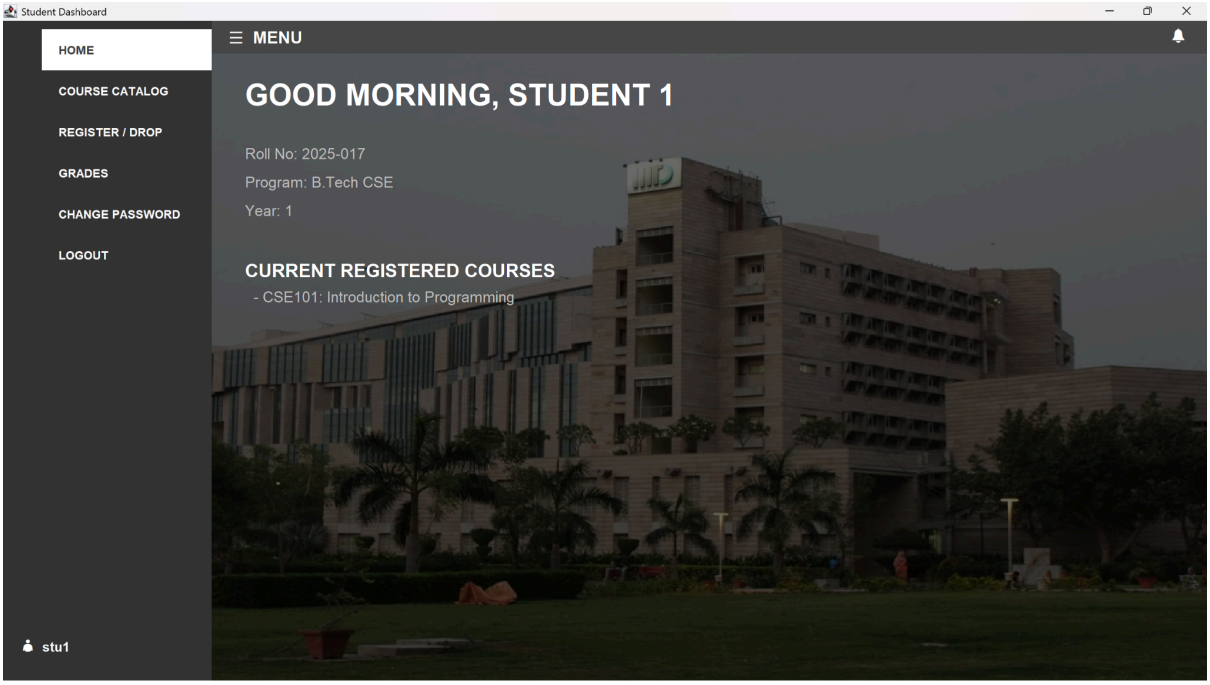

Admin Dashboard
HOME
ADD USERS
CREATE/EDIT COURSES
ASSIGN INSTRUCTOR
TOGGLE MAINTENANCE
BACKUP & RESTORE
SEND NOTIFICATIONS
VIEW TIMETABLES
CHANGE PASSWORD
LOGOUT
admin1
MENU
VIEW TIMETABLES
View Schedule For:
Student
Student 1 (2025-017)
Student 1 (2025-017)
Student 2 (2025-018)
Student 3 (2025-019)
Student 4 (2025-020)
Student 5 (2025-021)
Student 6 (2025-022)
Student 7 (2025-023)
Student 8 (2025-024)
TIME
MON
THR
FRI
SAT
08:00 - 09:00
09:00 - 10:00
CSE101 (C01)
10:00 - 11:00
11:00 - 12:00
12:00 - 13:00
13:00 - 14:00
14:00 - 15:00
15:00 - 16:00
16:00 - 17:00

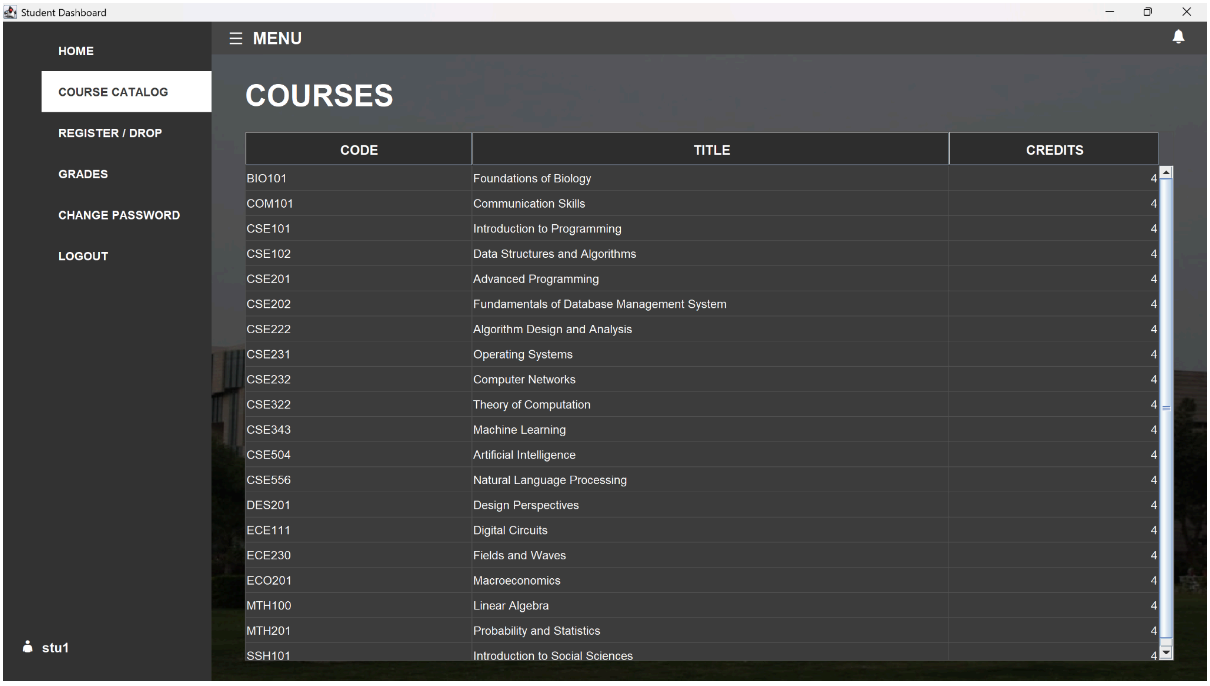


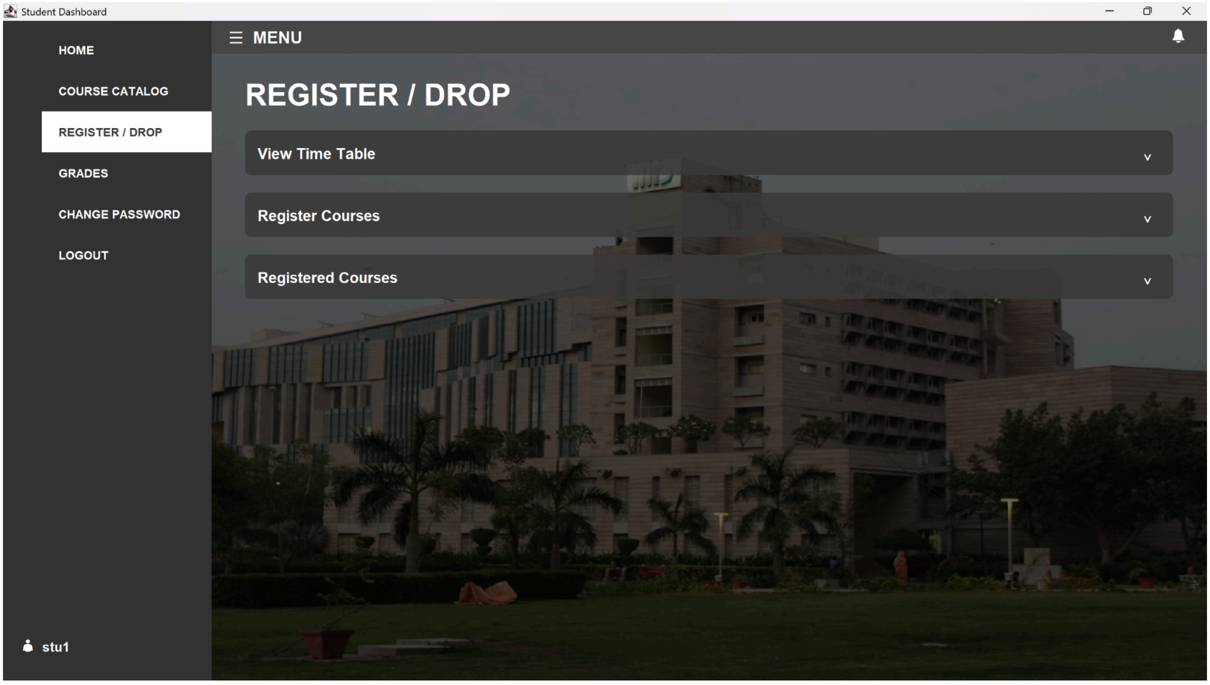
Login Window

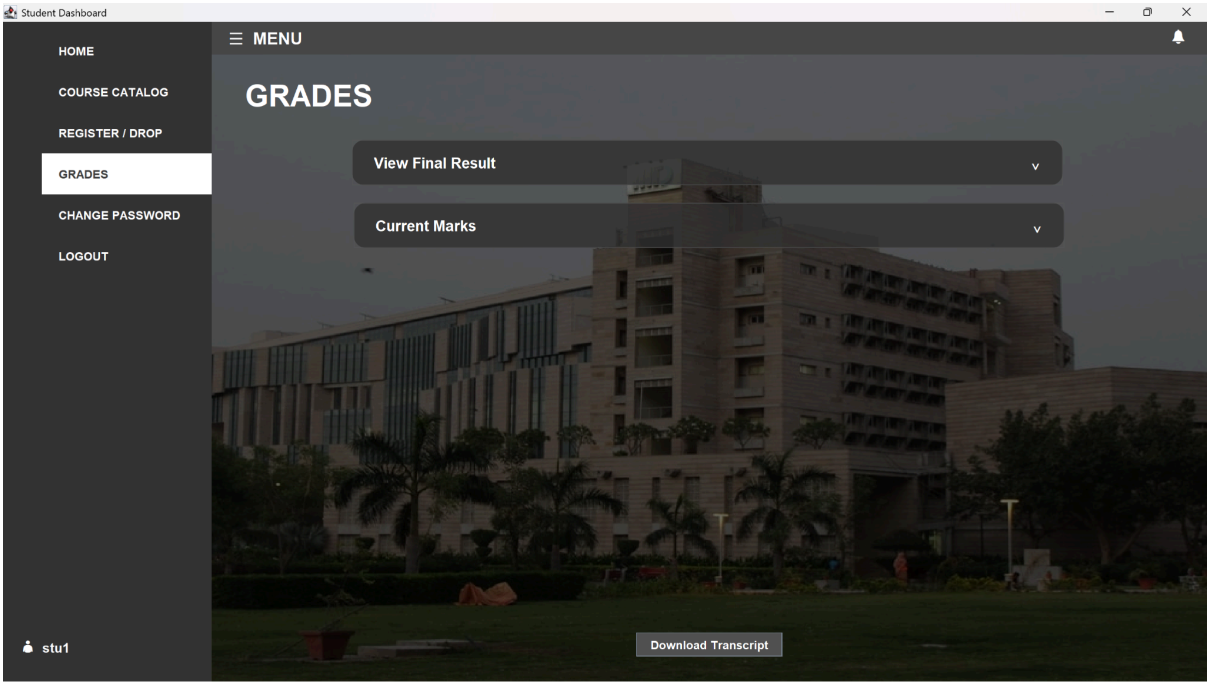


Student

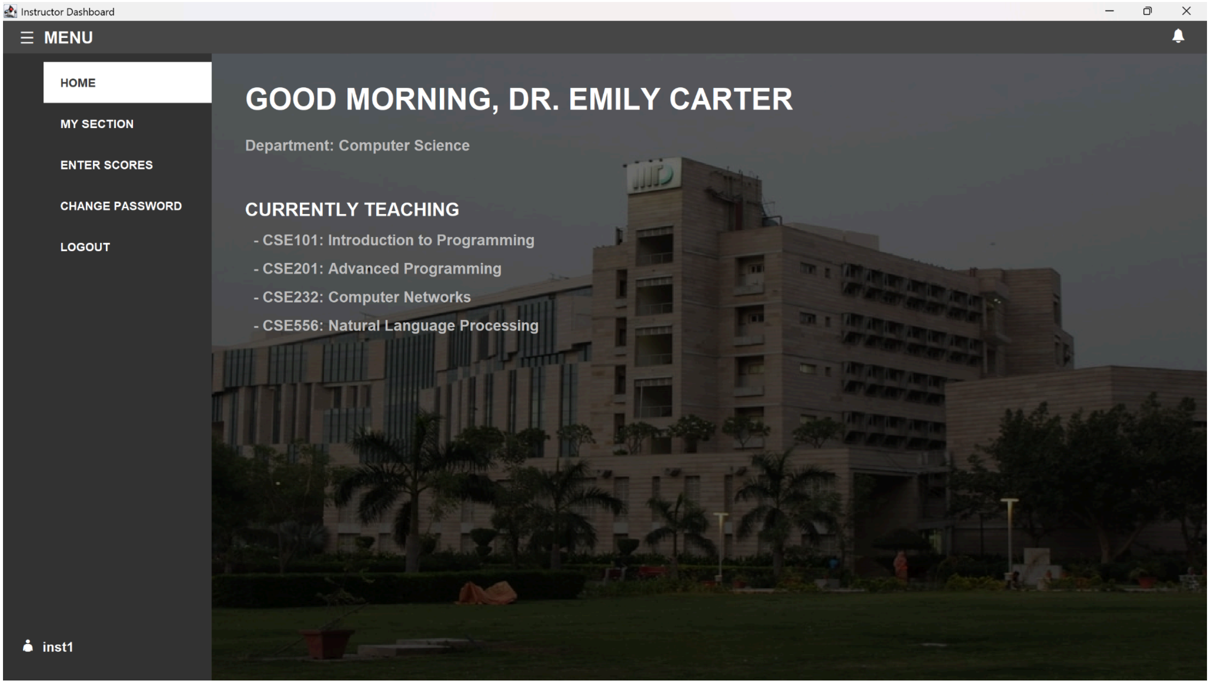

Student Dashboard
HOME
COURSE CATALOG
REGISTER / DROP
GRADES
CHANGE PASSWORD
LOGOUT
MENU
GOOD MORNING, STUDENT 1
Roll No: 2025-017
Program: B.Tech CSE
Year: 1
CURRENT REGISTERED COURSES
- CSE101: Introduction to Programming
stu1

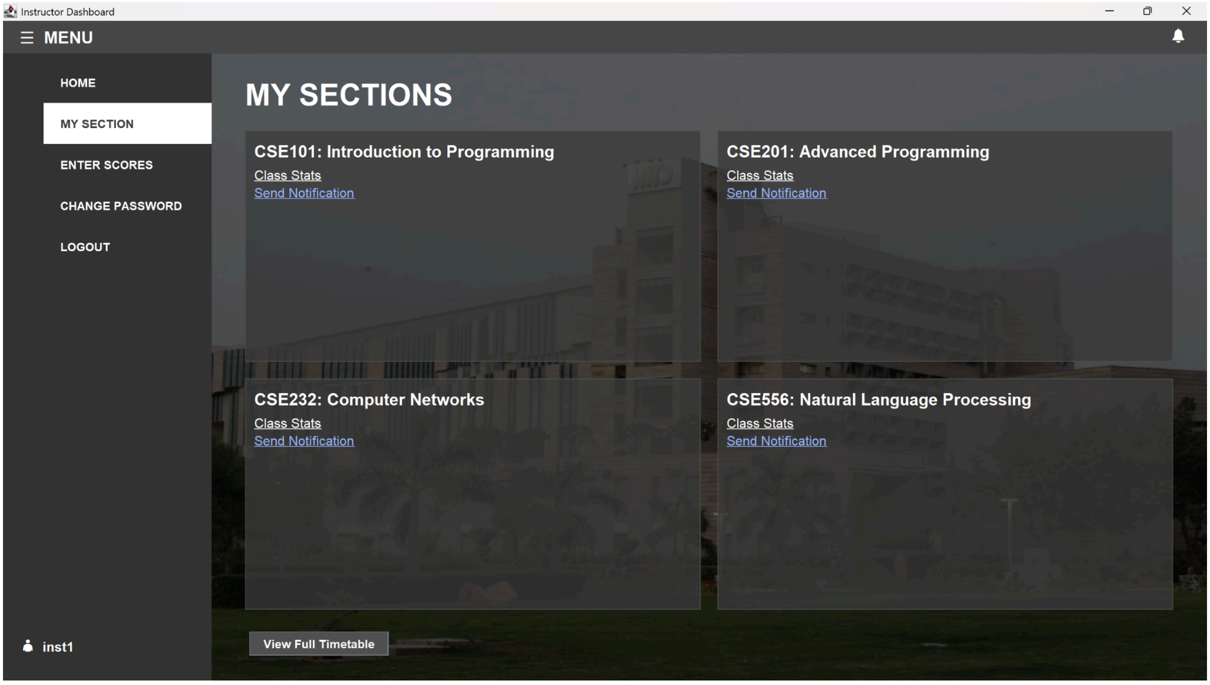

Student Dashboard
HOME
COURSE CATALOG
REGISTER / DROP
GRADES
CHANGE PASSWORD
LOGOUT
MENU
COURSES
CODE
TITLE
CREDITS
BIO101 Foundations of Biology 4
COM101 Communication Skills 4
CSE101 Introduction to Programming 4
CSE102 Data Structures and Algorithms 4
CSE201 Advanced Programming 4
CSE202 Fundamentals of Database Management System 4
CSE222 Algorithm Design and Analysis 4
CSE231 Operating Systems 4
CSE232 Computer Networks 4
CSE322 Theory of Computation 4
CSE343 Machine Learning 4
CSE504 Artificial Intelligence 4
CSE556 Natural Language Processing 4
DES201 Design Perspectives 4
ECE111 Digital Circuits 4
ECE230 Fields and Waves 4
ECO201 Macroeconomics 4
MTH100 Linear Algebra 4
MTH201 Probability and Statistics 4
SSH101 Introduction to Social Sciences 4
stu1

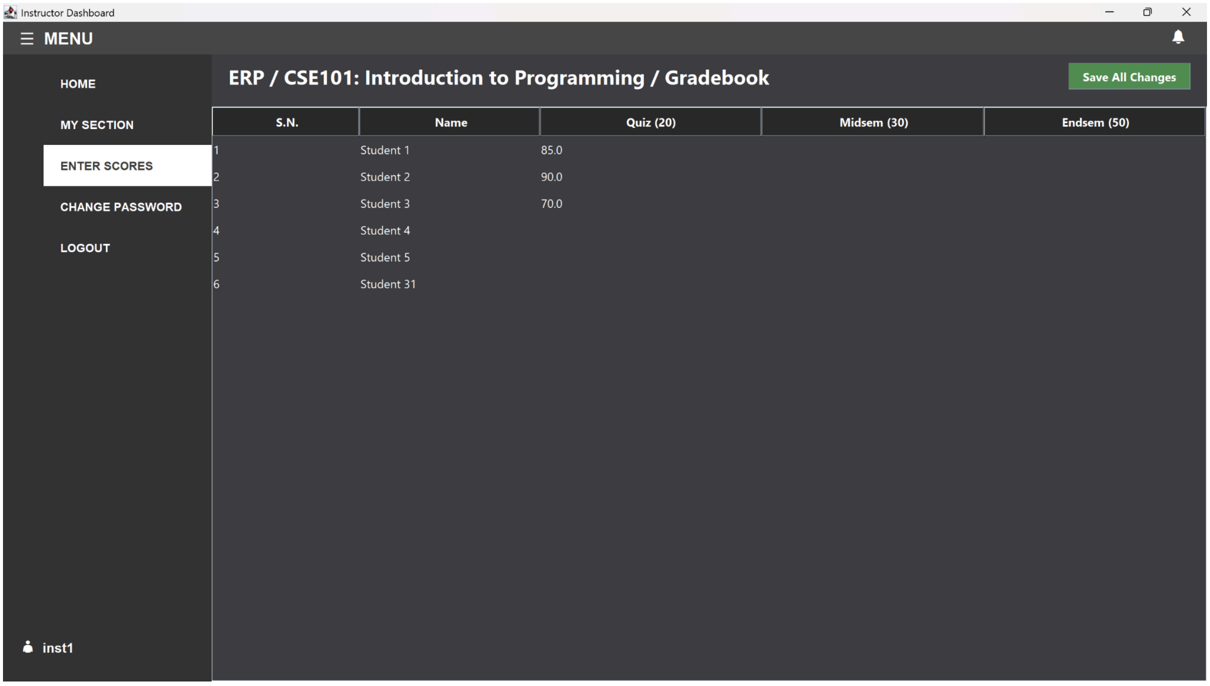

Student Dashboard
HOME
COURSE CATALOG
REGISTER / DROP
GRADES
CHANGE PASSWORD
LOGOUT
stu1
MENU
REGISTER / DROP
View Time Table
Register Courses
Registered Courses


Student Dashboard
HOME
COURSE CATALOG
REGISTER / DROP
GRADES
CHANGE PASSWORD
LOGOUT
stu1
MENU
GRADES
View Final Result
Current Marks
Download Transcript

Instructor


Instructor Dashboard
MENU
HOME
MY SECTION
ENTER SCORES
CHANGE PASSWORD
LOGOUT
inst1
GOOD MORNING, DR. EMILY CARTER
Department: Computer Science
CURRENTLY TEACHING
- CSE101: Introduction to Programming
- CSE201: Advanced Programming
- CSE232: Computer Networks
- CSE556: Natural Language Processing


Instructor Dashboard
MENU
HOME
MY SECTION
ENTER SCORES
CHANGE PASSWORD
LOGOUT
inst1
MY SECTIONS
CSE101: Introduction to Programming
Class Stats
Send Notification
CSE201: Advanced Programming
Class Stats
Send Notification
CSE232: Computer Networks
Class Stats
Send Notification
CSE556: Natural Language Processing
Class Stats
Send Notification
View Full Timetable


Instructor Dashboard
MENU
HOME
MY SECTION
ENTER SCORES
CHANGE PASSWORD
LOGOUT
inst1
ERP / CSE101: Introduction to Programming / Gradebook
Save All Changes
S.N.
Name
Quiz (20)
Midsem (30)
Endsem (50)
1 Student 1 85.0
2 Student 2 90.0
3 Student 3 70.0
4 Student 4
5 Student 5
6 Student 31